Dams mark drought vulnerability rather than buffer it, and storage cannot offset a falling water supply in Chile

Francisco Zambrano Bigiarini[a,*]

[a] *Universidad de Las Américas, Facultad de Medicina Veterinaria y Agronomía, Santiago, Chile,*

Abstract

Expanding reservoir storage is the reflexive response to intensifying drought. Yet whether dams build resilience or merely mark where vulnerability already concentrates has remained untested under sustained drying, for want of a credible counterfactual. Here we treat reservoir presence as a basin-level intervention across central Chile during its ongoing megadrought, using statistical matching to compare each dammed basin against undammed basins that share its climate, size, and terrain, and asking whether the two respond differently to the same drought. Across every operational pathway, meteorological-to-streamflow drought transmission, evapotranspiration, and irrigated-area change, the reservoir effect brackets zero once siting is matched out. The cleanest test compares regulated reaches below each dam against physically unregulated reaches above it: the apparent buffering is just as strong upstream as downstream and disappears when basins are compared within the same climate, the signature of an aridity gradient rather than of regulation. Positive controls confirm the design would have detected a real buffering effect had one existed. Meanwhile the seasonal refill stays efficient even as the whole storage band drifts downward, indicating a falling water supply rather than a degraded buffer. Reservoirs in Chile thus mark pre-existing, location-determined vulnerability rather than create resilience. Under structural aridification, expanding storage is a supply-blind response to a supply problem; adaptation must shift toward managing demand, reallocating water, and recharging aquifers.

Keywords: reservoir effect, drought propagation, socio-hydrology, causal inference, megadrought, Chile

1. Introduction

As the world dries, governments instinctively build more dams. Reservoirs are the dominant structural response to drought, and in the short term they work: more storage raises water availability and lowers the

*Corresponding author
Email address: frzambra@gmail.com (Francisco Zambrano Bigiarini)

frequency, severity and duration of shortage (Di Baldassarre et al., 2018). Yet a socio-hydrological view warns that the same infrastructure can be self-defeating over a dam's planning horizon. Greater supply invites agricultural, urban and industrial expansion, so demand grows to meet and then exceed each new increment of storage (the supply-demand cycle); and prolonged reliance on abundant stored water erodes the incentive to adapt, deepening dependence and magnifying the damage when a drought finally outlasts the reservoir (the reservoir effect) (Di Baldassarre et al., 2018). Consistent with this, global water demand has outpaced storage for decades, the storage returns of new dams are diminishing worldwide (Li et al., 2023), and most reservoirs now fail to refill under intensifying drought (Wang et al., 2024). Whether these feedbacks dominate, or are outweighed by the protection dams provide, remains debated (Kellner, 2021).

The obstacle is that the reservoir effect for drought has been described far more than it has been measured. Reservoirs are not sited at random: they are built where irrigation demand and aridity already concentrate, so dammed basins differ systematically from undammed ones before any drought strikes. The short-term hydrological benefit is documented, as regulation shortens downstream hydrological drought and lengthens the lag from meteorological drought (Wu et al., 2017, 2018), though it can also redistribute scarcity in space and time (López-Moreno et al., 2009). But the long-term, demand side is far less constrained, and separating a genuine reservoir effect from the siting that accompanies it requires a defensible counterfactual, namely what a drought's impact *would have been* in a basin had the reservoir not been built (Di Baldassarre et al., 2018). Prior dammed-versus-undammed and before-after comparisons cannot supply it.

Chile is an acute testbed. Since 2010 its central regions have endured a *megadrought*, an uninterrupted run of dry years now shading into structural aridification as precipitation falls and atmospheric evaporative demand rises (Garreaud et al., 2025, 2017; Zambrano et al., 2025). Because the 1981 Water Code makes water-use rights perpetual private property that the state cannot readily curtail during scarcity, the policy reflex has been supply-side: a national programme of new dams and inter-basin transfers, alongside market incentives that concentrated water in high-value, water-demanding export crops (Barría et al., 2021; Reade Malagueño and D'Odorico, 2024). Chile thus exhibits, in compressed form, every ingredient of the reservoir effect: an extreme and persistent forcing, a supply-side response, and demand expansion under weak adaptive feedback.

Here we ask whether reservoirs build resilience or merely mark, and postpone, vulnerability under prolonged drought. We treat the presence of a major reservoir as a basin-level intervention and construct the missing counterfactual in two steps. First, we compare each dammed basin only against undammed basins that share its climate and landscape, and condition on the drought actually experienced, asking whether dammed and comparable undammed basins convert the same meteorological deficit into different impacts. Second, within each dammed basin we compare regulated reaches below the dam against physically unregulated reaches above it, which share climate, geology and siting but not the reservoir. Together these turn the socio-hydrological reservoir-effect hypothesis into a falsifiable, causally framed test, in the setting where it matters most.

2. Methods

2.1. Study area and design

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyper-arid, Mediterranean, and humid-temperate climates, over 2005–2024, the window in which reservoir storage, meteorological forcing, and satellite outcomes jointly exist (`config/study_period.yml`). The design is a matched comparison of **dammed versus undammed sub-watersheds**, with reservoir presence treated as a basin-level intervention. Because 18 of 24 monitored reservoirs were commissioned before the storage record begins, treatment is effectively time-invariant over the window and a staggered-adoption (commissioning) event study is infeasible; identification therefore rests on a matched cross-section combined with conditioning on meteorological forcing rather than on calendar time.

The central estimand is the operational effect of a reservoir on drought outcomes *conditional on the covariates that govern where reservoirs are sited*, that is, whether dammed basins respond to the same drought forcing differently from hydroclimatically comparable undammed basins. This isolates reservoir *operation* from reservoir *siting* (Di Baldassarre et al., 2018).

2.2. Spatial units and treatment assignment

The analysis grain is the DGA BNA **sub-watershed (subcuenca)** (467 units), chosen over the coarser cuenca grain (101 units) for power and common support: reservoirs sit in Chile's largest main-stem basins, and the finer grain supplies far more controls inside the treated size range (`docs/design/grain-selection.md`). A subcuenca is **treated** if it contains 1 of the 26 monitored reservoirs by point-in-polygon assignment, giving **24 treated units**. An untreated subcuenca that lies within a cuenca containing a reservoir is flagged as **contaminated** (it shares the regulated main stem and is not a clean counterfactual) and removed (45 units), leaving **398 clean controls**. This contamination rule defuses the most direct downstream-spillover (SUTVA) violation; residual adjacency and shared-aquifer spillovers between neighbouring subcuencas are carried as an identification caveat and checked at cuenca grain.

2.3. Data

Reservoir storage and treatment. Monthly storage for 26 monitored reservoirs (2005-01 to 2026-04) with a per-reservoir capacity column (`max_level_hm3`); storage and capacity share units (hm^3), so percent-of-capacity (`100 × level / max_level_hm3`) is the cross-reservoir-comparable state variable. Static attributes (commissioning year, use, size class) come from the national dam registry.

Meteorological forcing. Standardized drought indices (SPI, SPEI, and EDDI) derived from CHIRPS precipitation and CHIRTS temperature at timescales of 1/3/6/12/24/36 months, standardized on the 1991–2026 baseline (`config/variables.yml`). SPEI (Vicente-Serrano et al., 2010) at 12 months is the primary exogenous dose; the suitability of CHIRPS-derived standardized indices for Chilean drought monitoring is established, including ENSO-modulated drought variability across the country (Meza, 2013; Oertel et al., 2020; Zambrano et al., 2017). We use the 12-month accumulation because it matches the annual, snowmelt-fed hydrological cycle of central Chile (winter precipitation stored as snow and released through the growing season) and so captures the inter-seasonal carryover that irrigation reservoirs are built to bridge, at the scale at which multi-year vulnerability (rather than sub-seasonal operation) is expressed. By construction this does not resolve short-term (1–3 month) operational buffering, which reservoirs are documented to provide (Wu et al., 2017, 2018) and which we flag as a boundary of our conclusions (Discussion). Conditioning on forcing (estimating effects on the deficit-to-impact *slope* rather than on levels) is how we break the collinearity between the 2010-onset megadrought and the treatment clock.

Hydrological drought. Daily streamflow from the CR2 network (2000–2020) is aggregated to the Standardized Streamflow Index at 12 months [SSI-12; Vicente-Serrano et al. (2012); Lema et al. (2025)], the regulated-flow analogue of SPEI-12, used as the outcome in the within-basin placebo (below). Gauges are classified as upstream or downstream of each dam from an SRTM digital elevation model.

Ecological and land-use outcomes. (i) annual irrigated/cropland area from MapBiomas Chile Collection 2 (30 m, 1999–2024); (ii) actual evapotranspiration from MODIS (8-day, ~500 m, 2001–2024), for whole basins and for an orchard stratum from the CIREN/ODEPA cadastre; (iii) standardized vegetation and productivity anomalies, namely monthly z-score accumulated NDVI (zcNDVI-6), annual z-score NPP (zNPP), and the standardized actual-ET anomaly (SETI), from MODIS over 2000–2024; the zcNDVI productivity proxy is validated for Chilean agricultural drought, and the ecological response of Chilean vegetation to the megadrought is well characterized (Glade et al., 2016; Venegas-González et al., 2023; Zambrano et al., 2018).

Matching covariates. Baseline aridity (long-term mean annual P/PET over the 1991–2020 WMO normal), watershed area, mean elevation (SRTM), and Köppen–Geiger climate class, extracted per subcuenca by zonal statistics.

2.4. Matched control construction

We estimate the average treatment effect on the treated (ATT) using **entropy balancing** (`WeightIt`, `method = "ebal"`) exact-matched within Köppen main group, balancing **log(area)** and **mean elevation** (`docs/design/matched-controls.md`). Balancing on size and elevation rather than latitude is deliberate: latitude and the climate class fight each other across the Andes–Patagonia span, and substituting SRTM elevation raised the effective control sample size from 9 to 91 while balancing latitude for free.

The procedure retains **21 of 24 treated subcuencas** (three high-Andes units fall outside local elevation common support) and achieves essentially perfect balance (standardized mean differences for log(area) and elevation fall from 1.18 / 0.37 to 0.000, variance ratios 1), with an **effective control sample size 91** out of 244 in-window controls. Baseline aridity is a genuine confound (dammed basins are markedly more arid: median P/PET 0.26 vs 0.80) but is *not* used as a balance target, because hard-balancing it collapses the ESS from 91 to 19; instead the residual aridity imbalance (SMD 0.17, above the 0.1 target) is absorbed by doubly-robust regression adjustment. Because the matched set does not fully balance aridity, this adjustment is *load-bearing* rather than cosmetic, and the estimates can be sensitive to its functional form; we therefore stress-test the aridity-adjusted estimates with a quadratic aridity term and report the curvature-robust values where feasible (see Results). Robustness to the matching method is assessed with coarsened exact matching (CEM) and 1:2 nearest-neighbour (Mahalanobis) matching on the identical common-support sample; entropy balancing and NN agree on all 21 treated, while CEM additionally prunes the 4 most extreme large/high basins, so all treatment effects are reported with and without those units.

2.5. Causal estimands and estimators

Doubly-robust matched ATT. Each outcome is estimated three ways on the same sample, comprising weighting-only (ebal-weighted difference in means), regression-only (covariate-adjusted g-computation), and the headline **doubly-robust** estimator (ebal weights *and* an outcome model in log(area), elevation, and log(aridity)), so functional-form sensitivity is visible. Continuous covariates are centred at the treated mean so the treatment coefficient reads as the ATT; standard errors are HC3-robust (ebal weights treated as fixed, the standard mildly anti-conservative simplification).

Forcing-conditioned transmission slope. To remove megadrought *exposure* (concentrated in exactly the arid basins where reservoirs sit), the outcome for the DR machinery is the per-basin **transmission slope**, the regression of an annual outcome (e.g. zNPP) on annual SPEI-12 over 2005–2024, a standardized, cross-basin drought-transmission elasticity. The vulnerability hypothesis predicts a *steeper* slope (more impact per unit deficit) in dammed basins.

Forcing-interacted difference-in-differences. On the observed area and ET panels we fit

$$y_{it} = \beta\,(\text{dammed}_i \times \text{SPEI}_{it}) + \gamma\,\text{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with subcuenca (α_i) and year (τ_t) fixed effects and entropy-balancing weights. Because treatment is time-invariant, the dose is meteorological SPEI rather than calendar year: the year fixed effects absorb the common megadrought shock, and β is the differential deficit-to-impact slope between dammed and matched-control basins, robust to the siting-in-arid-central-Chile confound, which biases levels far more than slopes. The vulnerability hypothesis predicts $\beta > 0$.

Equivalence testing and minimum detectable effect. Because the headline results are nulls, we move beyond failure-to-reject and bound the operational effect. For each null we report a two one-sided test (TOST): the effect is declared equivalent to zero if its 90% confidence interval lies within a negligible region of ±25% of the baseline (untreated) transmission slope. We are not aware of an established hydrological or operational standard defining a "negligible" reservoir effect on drought transmission, so we anchor the margin in the granularity of operational drought classification rather than choosing it arbitrarily. Standardized drought indices are managed in discrete severity categories each 0.5 standardized units wide (McKee et al., 1993; Vicente-Serrano et al., 2010); over the observed range of meteorological deficits a change of <25% in the deficit-to-impact slope (0.13 SSI units on the 0.53 baseline) alters the propagated streamflow-drought index by less than a single severity category, i.e. it cannot move a basin across an operational drought-severity threshold and is therefore marginal for drought management. We adopt ±25% as this deliberately strict, pre-specified margin and, because the equivalence verdict is nonetheless sensitive to the bound, also report the threshold-free MDE so that readers may apply any margin they consider material. The qualitative conclusion (only large effects are excluded; no outcome is equivalent at ±25%) does not depend on the exact margin. The minimum detectable effect (at 80% power, two-sided $\alpha = 0.05$) and the equivalence interval are built from the **standard deviation of the permutation-null distribution**, not the analytic standard error, since cluster-robust SEs over-reject at ~21 clusters (`src/R/causal/equivalence.R`). To confirm the null is informative rather than underpowered, we additionally run a **positive control**: a known buffering slope is injected into the treated units and the full estimator and randomization inference are re-run, verifying that the design recovers an effect of policy-relevant size and correctly flags none when none is injected.

Siting-confound decomposition. To quantify how much of any apparent buffering is selection, we fit a ladder on the same panel: (1) a naive pooled slope gap (no weights, no fixed effects, what a conventional dammed-vs-undammed study reports); (2) unit + time fixed effects, unweighted; (3) the design estimator (ebal weights + fixed effects). The shrinkage from (1) to (3) is the siting confound; the residual at (3) is the regulation estimate.

Within-basin upstream/downstream placebo. The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated **downstream** reaches against physically unregulated **upstream** reaches of the *same* dammed basins, which share climate, geology, and selection history. A genuine regulation effect implies a downstream-specific dose-response that the upstream placebo does not reproduce; an attenuation equally present upstream is siting, not regulation. Of the 21 treated matched basins, 15 have both an upstream and a downstream gauge, giving near-national coverage rather than a single case study. The contrast is also split by Köppen region to test whether a regulation effect emerges in the high-supply south. Because upstream gauges sit at higher elevations than downstream gauges, we test the placebo's comparability assumption directly: we quantify the elevation gap and regress the per-gauge transmission slope on elevation among undammed control gauges, so that any elevation-driven regime difference is measured rather than assumed.

2.6. Inference

With only ~21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is **randomization inference**: the treatment label is permuted among units within Köppen × aridity- tercile strata (999 permutations) and the observed statistic is compared against the resulting null distribution; cluster-robust confidence intervals are reported alongside for completeness. Parallel pre-trends are checked with a dynamic event study (dammed × year, reference 2009), and residual confounding is probed with an aridity placebo that relabels the most-arid control tercile as pseudo-treated.

2.7. *Software and reproducibility*

All analyses are implemented in R within a `targets` pipeline (matched-set construction, estimators, and figures are individual targets), using `WeightIt`/`cobalt` for balancing and diagnostics, `fixest` for fixed-effects regression, `sf`/`terra`/`exactextractr` for the spatial extractions, and `renv` for package pinning. Every result in the manuscript corresponds to a named pipeline target; no figure or table is produced by hand.

3. Results

3.1. *Reservoirs sit where water is already scarce*

Reservoirs in Chile are built where drought bites hardest: our 21 dammed basins cluster in the arid centre and north, where irrigation demand and aridity already coincide, so a raw comparison of dammed against undammed basins would confuse the reservoir with its location. We separate the two with the design in Figure 1. First, we compare each dammed basin only with undammed basins that share its climate, size and elevation, statistical matching that keeps 21 of 24 dammed basins against an effective 91 of 244 controls (Supplementary Figs. S4 and S5), and we condition on the drought each basin actually experienced, so we compare how much impact a basin registers per unit of meteorological deficit rather than its raw level. Second, within each dammed basin we compare regulated reaches below the dam against physically unregulated reaches above it. A genuine reservoir effect must show up in these comparisons; across every pathway, it does not.

3.2. *The apparent buffering is the location, not the dam*

The most direct and intuitive test is on streamflow. Meteorological drought (SPEI-12) propagates into streamflow drought (SSI-12) more gently in dammed basins, the visual signature of buffering (Figure 2, panel a). But this flattening tracks where dams sit, not what they do. If the dam were doing the buffering, transmission would flatten only below it; instead it flattens just as much at the unregulated *upstream* reaches of the same basins (differential slope -0.20 upstream versus -0.16 downstream; Figure 2, panel b). Because the upstream line flattens too, the cause must be the aridity of the location, not the reservoir. The national slope gap of -0.18 accordingly collapses to about -0.03 once dammed and control basins are compared within the same climate region, and in the high-supply south, where buffering should be easiest to see, the upstream and downstream gaps are statistically identical (-0.036 versus -0.034). None survives randomization inference ($p_{\text{perm}} = 0.39$). The apparent buffering is therefore the between-region aridity gradient, the textbook signature of a siting confound, not an effect of regulation; a step-by-step decomposition

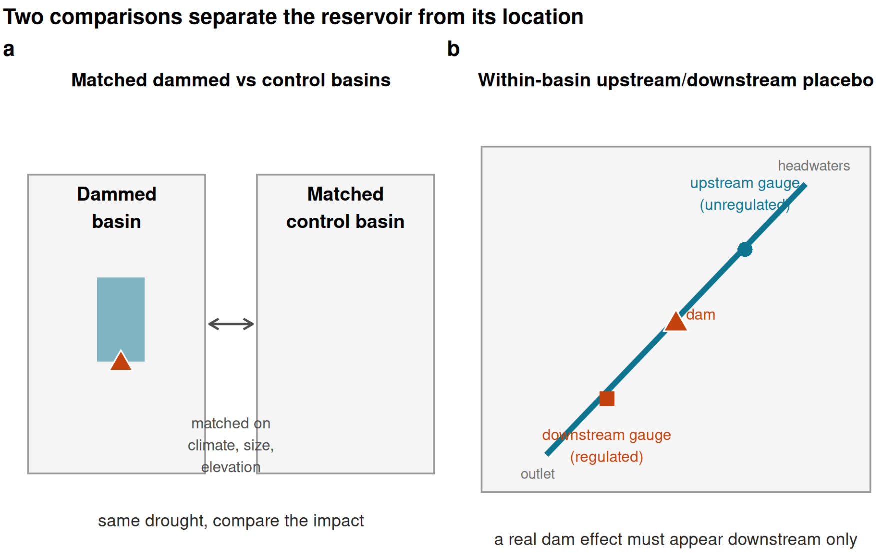

Figure 1: The two comparisons that separate the reservoir from its location. (a) Matched comparison: each dammed basin is compared only with undammed basins that share its climate, size and elevation, and both are then exposed to the same drought so their impact can be compared. (b) Within-basin placebo: inside a dammed basin, a streamflow gauge below the dam (regulated) is compared with a gauge above it (unregulated); a genuine reservoir effect must appear downstream only. Schematic; the full study-area map and matching diagnostics are in Supplementary Figs. S4 and S5.

confirms the signal survives matching and fixed effects but dies only when basins are compared within climate region (Supplementary Fig. S3 and Table S4). Upstream and downstream reaches are not perfectly interchangeable, since upstream gauges sit higher and elevation predicts transmission among controls, but their raw transmission slopes are nearly identical and the apparent buffering is if anything stronger upstream, neither of which is consistent with a regulation effect hidden by elevation (Supplementary Table S2).

3.3. No reservoir effect survives across any operational pathway

The same answer holds across the other pathways by which reservoirs are thought to modify drought (Figure 3, Table 1). We measure each effect as the differential response of dammed and matched-control basins to the same meteorological deficit, and place them on a common axis: each estimate is divided by its own standard error, so effects on different natural scales (hectares, evapotranspiration, streamflow) can be read on one plot, and a value inside ± 2 is indistinguishable from no effect. Every interpretable estimate brackets zero: cross-sectional irrigated-area expansion (-0.69 ha km^{-2}, 95% CI $[-2.4, 1.1]$) and evapotranspiration buffering (-0.014 $[-0.065, 0.037]$), and the forcing-conditioned slope gaps for irrigated area ($p_{\text{perm}} = 0.91$) and orchard evapotranspiration ($p_{\text{perm}} = 0.24$). One outcome, whole-basin evapotranspiration, appears positive, but its pre-trends fail and it is confounded by residual aridity over barren land, so we flag it as uninterpretable rather than fold it silently into the nulls; at face value it points toward *vulnerability*, not buffering.

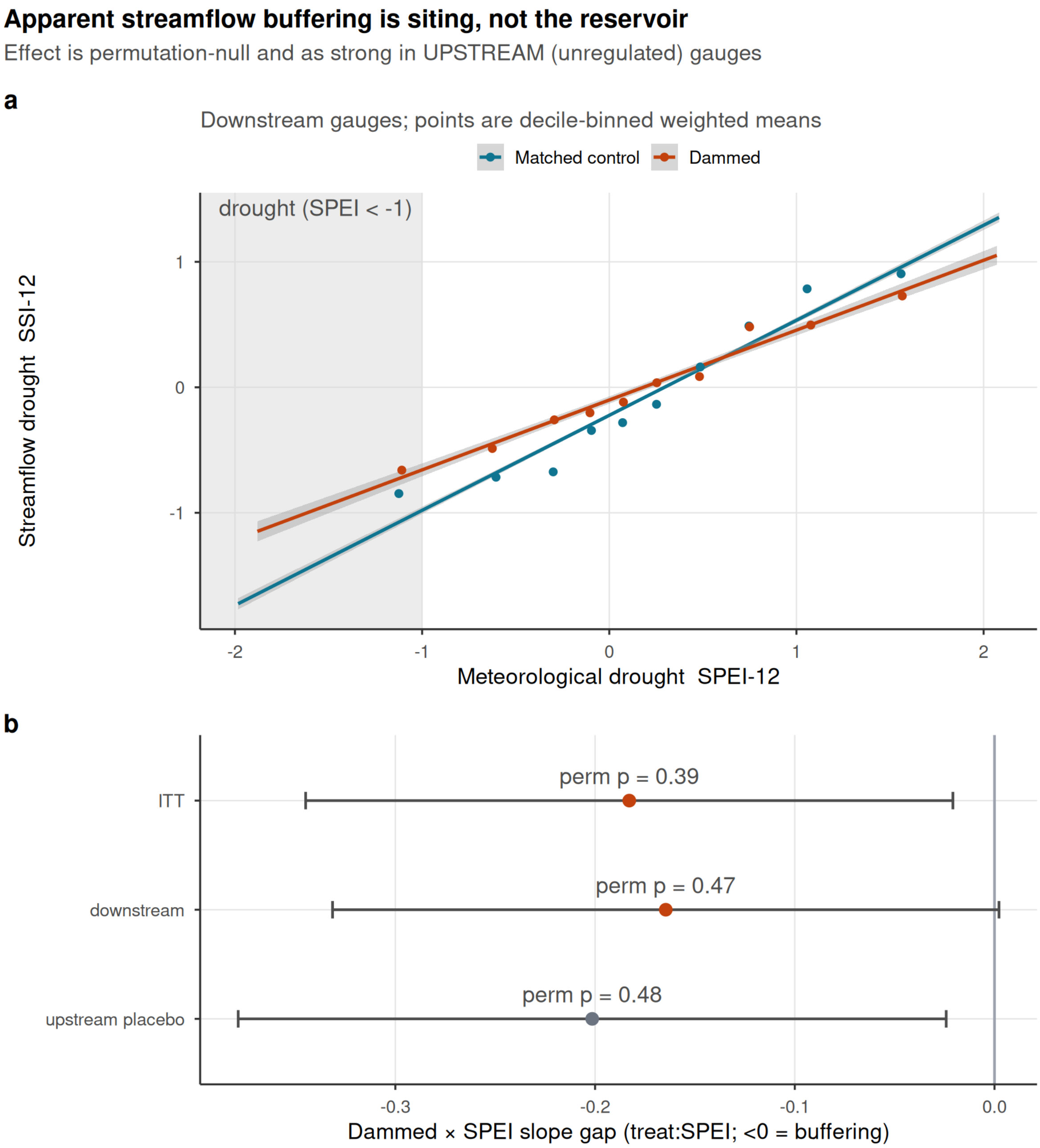

Figure 2: Streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); lines are weighted least-squares fits and shaded bands are 95% confidence intervals; dammed basins show a flatter slope, the apparent buffering. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo; horizontal bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE), annotated with randomization-inference p-values. All are permutation-null, and the upstream placebo (above-dam, unregulated) is as strong as downstream, so the attenuation is siting/aridity, not the dam.

This null is informative, not merely underpowered. With about 21 dammed clusters the design cannot claim exact equivalence to zero, but it rules out streamflow buffering larger than 46 to 58% of baseline transmission, and a positive control confirms the point: when we inject a known buffering effect into the treated gauges, the estimator recovers it one-to-one and randomization inference detects it once it exceeds that threshold (Methods; Supplementary Fig. S2 and Tables S1, S3). The design would therefore have detected a large operational effect had one existed.

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs and forcing-interacted DiD slope gaps (treat:SPEI). DiD rows are judged by randomization-inference p-values (p), the design's primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. *Whole-basin ET fails its event-study pre-trend test (p approximately 1e-10): its cluster-robust CI excludes zero but the panel is confounded by barren-land aridity, so it is labelled confounded and excluded from the equivalence bounds (Supplementary Table S1).

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Cross-sectional matched ATT	Irrigated-area expansion	ha km^-2 added	-0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Reservoir ET buffering	d log ET / d SPEI	-0.014	[-0.0646, 0.0369]	-	null
Forcing-interacted DiD (slope-gap)	Irrigated area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null
Forcing-interacted DiD (slope-gap)	Whole-basin ET	differential deficit->impact slope	0.048	[0.0255, 0.0703]	0.11	confounded*
Forcing-interacted DiD (slope-gap)	Orchard ET	differential deficit->impact slope	-0.020	[-0.0448, 0.00565]	0.24	null

3.4. *Reservoirs mark, not make, irrigated vulnerability*

Dammed basins do carry the vulnerability reservoirs are blamed for concentrating: they hold about 4.6 times more cropland than matched controls (11.1% versus 2.4% of basin area; Figure 4, panel a). But this gap is a *siting* signature, present and stable across the entire 2000–2024 record, with dammed and control trajectories evolving in parallel through the megadrought. A dynamic event study confirms it: the differential trend is flat before 2010 ($p = 0.65$) and does not diverge afterwards ($p = 0.81$; Figure 4, panel b). Reservoirs mark where irrigated agriculture already was; they neither expanded it nor changed how it responds to drought.

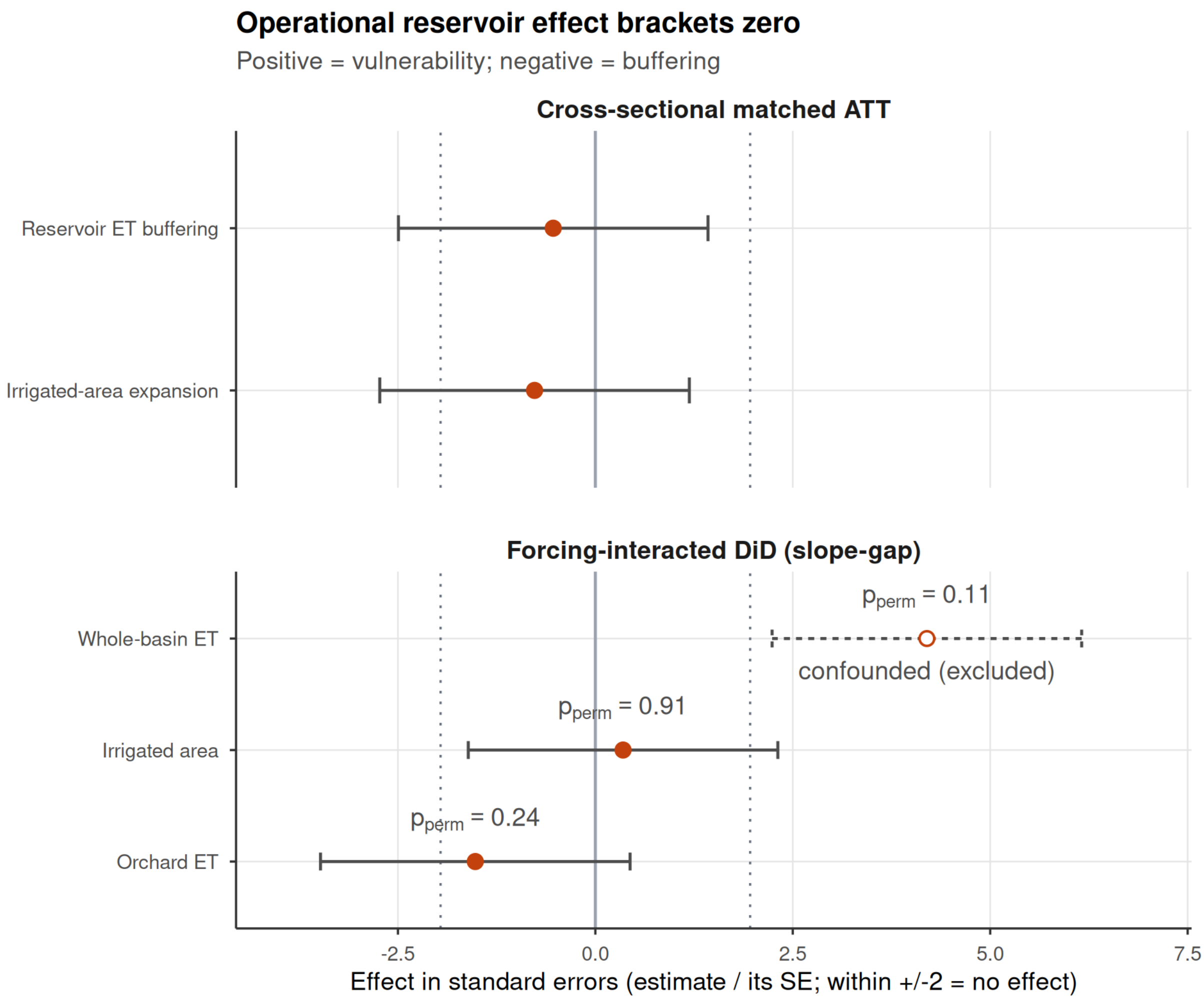

Figure 3: Dammed-versus-control effect across the estimators, each expressed uniformly in its own standard errors (estimate divided by its SE) so the axis metric is identical for every row; an estimate inside +/-2 is indistinguishable from no effect. Rows are grouped by inference regime: cross-sectional matched ATTs (top) and forcing-interacted DiD slope gaps (bottom). Bars span +/-1.96 SE (the 95% reference interval) and dotted lines mark +/-1.96; a positive effect would indicate induced-demand vulnerability, a negative one operational buffering. DiD rows are annotated with their randomization-inference p (the design's valid inference at ~21 clusters, since the +/-1.96 SE guides over-reject there). Every interpretable estimate brackets zero. Whole-basin ET (open symbol, dashed interval) is the sole exception: its estimate/SE exceeds +1.96 and its cluster-robust CI excludes zero, but the panel fails its event-study pre-trend test and is confounded by barren-land aridity, so it is labelled confounded and excluded (Table 1); taken at face value it points toward vulnerability, not buffering.

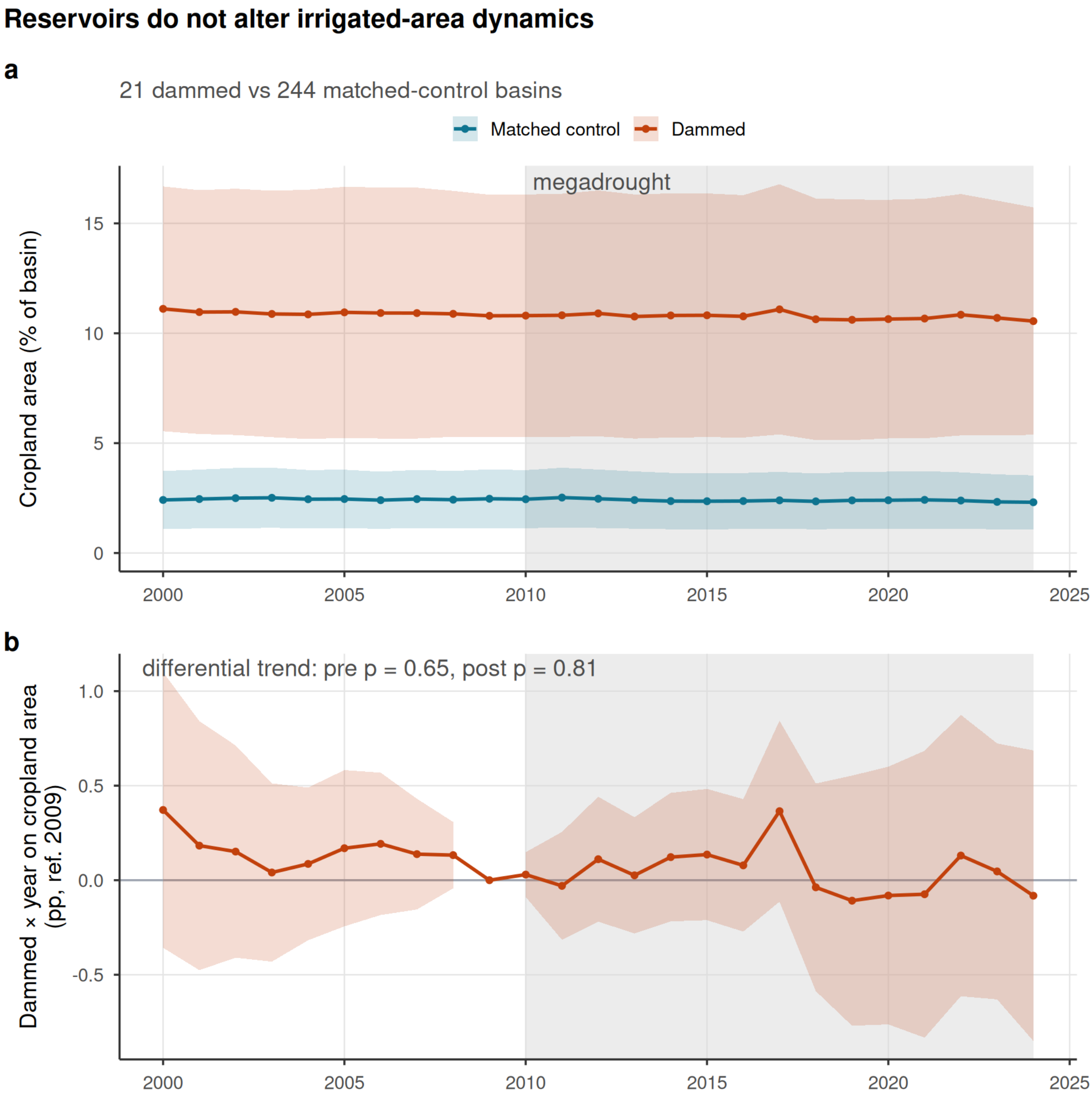

Figure 4: Observed irrigated-area dynamics in dammed versus matched-control basins (21 dammed, 244 controls). (a) Entropy-balancing-weighted mean cropland area (% of basin, MapBiomas class 18) with 95% bands (weighted mean +/- 1.96 SE across basins); the y-axis is anchored at zero. Dammed basins hold ~4.6x more cropland (11.1% vs 2.4% of basin area) but the trajectories evolve in parallel. (b) Dynamic event study (dammed x year, reference 2009): the shaded band is the 95% confidence interval (coefficient +/- 1.96 cluster-robust SE). The differential pre-trend is flat (pre-2010 trend-slope $p = 0.65$) and does not diverge afterwards (post $p = 0.81$); a linear trend-slope test is used because the joint event-study Wald over-rejects at ~21 treated clusters. The 2010-2024 megadrought is shaded in both panels.

3.5. *Storage falls while the seasonal refill holds*

If reservoirs neither buffer nor amplify drought, the change that matters under the megadrought is in how much water there is to store (Figure 5). Over 2005–2024 the entire storage band drifts downward in step: annual peak storage falls by 1.3 percentage points of capacity per year ($p = 0.013$) and the trough almost identically (1.2 pp yr^{-1}, $p < 10^{-4}$), while the seasonal amplitude is statistically unchanged (-0.06 pp yr^{-1}, $p = 0.91$; Supplementary Table S5). A band that drops without narrowing means the reservoirs still refill efficiently each year but from a shrinking supply, consistent with a falling water-supply ceiling rather than a degraded buffer. This trend is descriptive, with no control group or naturalized-inflow series, but our most direct demand proxy weighs against a demand-driven explanation: irrigated area in dammed basins did not expand (Figure 4, Table 1), so a growing consumptive footprint is not evident as the driver (Li et al., 2023; Wang et al., 2024).

4. Discussion

Across a matched national design and four independent estimators, and on the direct streamflow-availability outcome, the operational effect of reservoirs on drought brackets zero once siting is matched out. Dammed basins do hold about 4.6 times more cropland than matched controls, but that gap is fixed across the record, with parallel pre-trends and no divergence through the megadrought. The evidence supports a single reading: reservoirs **mark** where water-dependent vulnerability already concentrated, in arid, already-irrigated basins, rather than **make** it. The vulnerability signature is a property of siting, set before the drought and stable through it, not a dynamic consequence of regulation.

This is a finding, not a failure of power, because a credible *cause* of the apparent effect is identified. Our result of no operational effect is not blindness but a refutation of a core policy assumption: if dams did what they are built to do, our design was powerful enough to detect it, as the positive controls demonstrate, so their failure to do so is itself informative and means policymakers must look elsewhere. The within-basin upstream/downstream placebo is the cleanest falsification: a genuine regulation effect must appear below the dam and be absent above it, and it is not. What remains, once the placebo and within-region conditioning are applied, is the aridity gradient itself. The methodological corollary is general and transferable: a naive dammed-versus-control contrast, or a calendar-time difference-in-differences, reads the large cross-sectional gaps (amplified during drought) as a reservoir effect, whereas conditioning on realized forcing and benchmarking against an unregulated within-basin reach makes the signal disappear. Reservoir-buffering

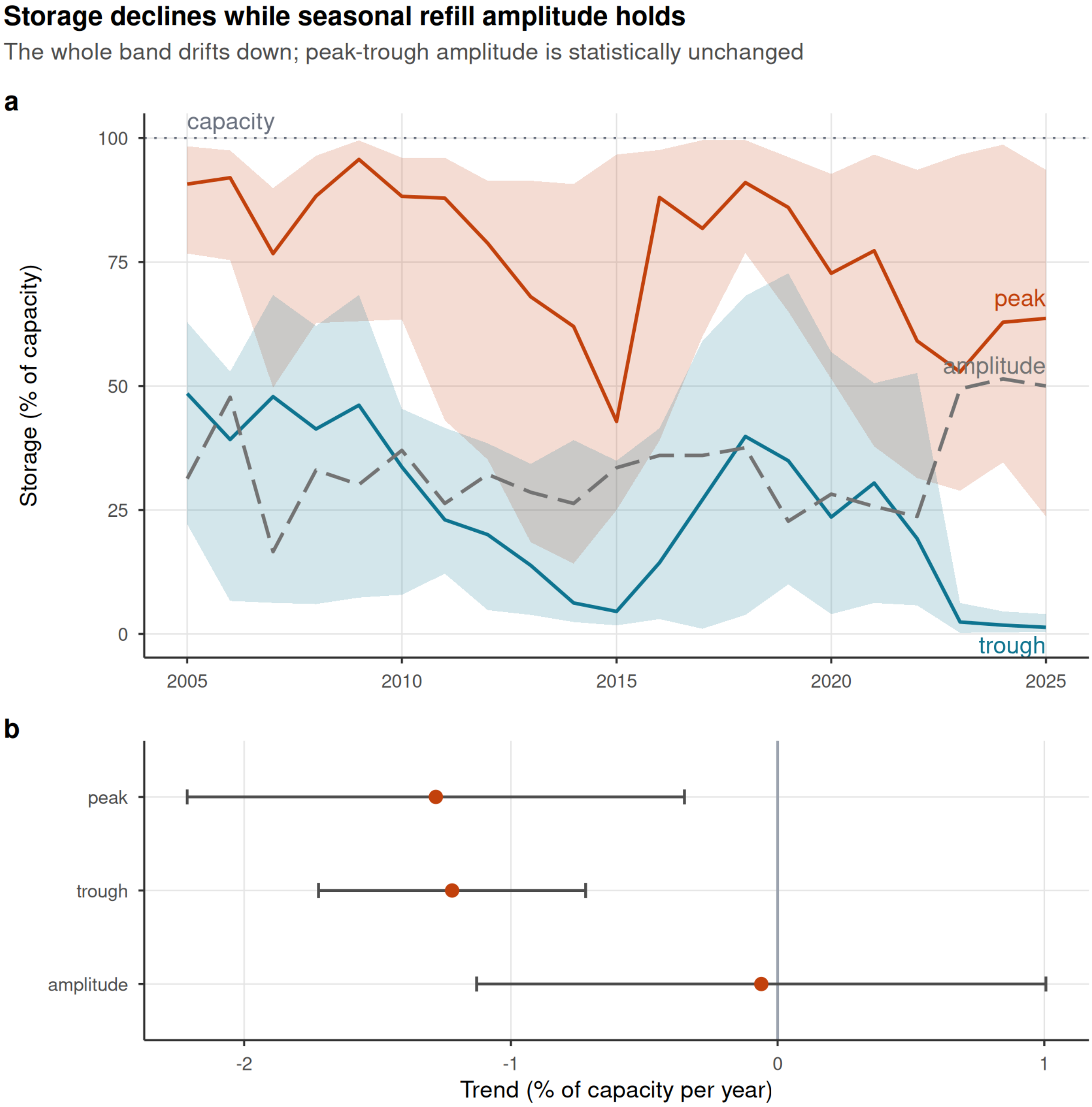

Figure 5: Storage declines while seasonal refill amplitude holds. (a) Cross-reservoir annual storage band: realized median peak and trough percent-of-capacity (solid lines) with interquartile ribbons across reservoirs, and the peak-trough amplitude (grey dashed). Medians are used because a few reservoirs report storage above nominal capacity (dotted 100% line); the whole band drifts down over 2005-2024 while the amplitude does not narrow. Central lines are the realized medians, not linear fits; the net trend is quantified in (b). (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects); error bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE): peak and trough both decline while the amplitude trend is not distinguishable from zero (a wide interval, so statistically unchanged rather than proven constant), consistent with a supply-side level decline rather than refill/buffer degradation. Descriptive: pooled within-reservoir trend, no control group.

studies that compare dammed against undammed basins, or pre- against post-construction levels, inherit the full siting confound (Lee et al., 2026; López-Moreno et al., 2009; Schilstra et al., 2024; Wu et al., 2017); the upstream placebo applies to any regulated river with gauges above and below the structure.

If reservoirs neither buffer nor amplify drought operationally, the change that matters under a multi-decadal dry spell is in the water available to store. The storage record is suggestive: the seasonal refill remains efficient and its amplitude unchanged, yet the whole storage band drifts downward, peak and trough declining in step at roughly 1.3 and 1.2 percentage points of capacity per year. A band that drops without narrowing points to a falling supply ceiling rather than a degraded buffer. This is a descriptive, control-free trend that cannot by itself separate declining inflow from rising demand (Condon et al., 2020), but our most direct demand proxy weighs against the demand explanation, since irrigated area in dammed basins did not expand. Read against Chile's documented transition from episodic drought toward structural aridification (Alvarez-Garreton et al., 2021; Garreaud et al., 2017; Vicente-Serrano et al., 2022; Zambrano et al., 2025) and the global pattern of reservoirs failing to refill and diminishing storage returns from new dams (Li et al., 2023; Wang et al., 2024), the more likely reading is that the binding constraint is shifting toward water supply rather than storage capacity or operation.

These findings speak to three international debates. They are the drought-domain expression of the socio-hydrological *reservoir effect* and *supply-demand cycle*, whose practical force has been contested for want of quantification (Di Baldassarre et al., 2018; Kellner, 2021): we provide a credibly identified test and find the protective function over-attributed to storage. They parallel the *efficiency paradox* in irrigation, in which measures that nominally save water expand consumptive use rather than reduce it (Grafton et al., 2018). And they align with the global evidence that the marginal returns of new dam construction are falling as reservoirs increasingly fail to refill (Li et al., 2023; Wang et al., 2024). In each case the lesson is the same: adding supply-side infrastructure does not resolve a scarcity that is driven by demand or by a falling supply.

The policy implication is direct and uncomfortable. Because Chile's 1981 Water Code makes water-use rights perpetual private property the state cannot readily curtail during drought, the reflexive response has been supply-side, a national programme of new dams and inter-basin transfers (Barría et al., 2021; Garreaud et al., 2025; Macpherson et al., 2023). For Chile and other regions facing structural aridification, our evidence indicates that building more storage is a supply-blind response to a supply problem: it cannot manufacture the water that aridification is removing, and the buffering benefit dams are built to provide is, on this evidence, a property of where they sit rather than of what they do. Adaptation must shift toward demand management, reallocation, and managed aquifer recharge rather than new impoundments (Grafton et al., 2018; Rivera-Vidal et al., 2025; Scanlon et al., 2023).

Several limitations bound these conclusions. Treatment is effectively time-invariant (most dams predate the record), so we identify differential drought *response* rather than the effect of switching a reservoir on; a staggered-commissioning design would be cleaner where construction dates and longer records permit. The result is a bounded null, not a proven equivalence: with ~21 clusters the design excludes only *large* buffering (>46–58% of baseline transmission for streamflow), so the causal weight rests on the placebo (Section 3.2), not on equivalence (Section 3.3). The observed-area outcome captures total cropland, not irrigation status, so a compositional shift toward water-intensive perennials could raise consumptive use without enlarging area; and the 12-month aggregation targets annual-to-interannual transmission, not the sub-seasonal buffering reservoirs also provide, so our conclusions concern long-term vulnerability rather than short-term operation. Finally, the operational finding, that buffering reflects siting not regulation, should hold wherever dam siting is non-random, but the demand-feedback and policy inferences are conditioned on Chile's unusually market-based, weakly adaptive governance; under state-managed allocation, operators able to curtail demand or re-time releases could in principle produce genuine buffering. A remaining possibility we cannot fully adjudicate is that damming leaves the magnitude of impact unchanged while recomposing its *dependence structure*, substituting a storage-to-outcome coupling for the meteorology-to-outcome one; resolving this needs reservoir-release and rule-curve records that set storage independently of contemporaneous inflow, and any such decoupling must still survive the upstream placebo.

Returning to the question that motivates the study, whether reservoirs create resilience or merely postpone vulnerability under prolonged drought, our evidence indicates that under aridification they do neither operationally: storage neither buffers nor amplifies the drought signal beyond what siting already implies, while the water available to store steadily declines. Resilience to a falling supply ceiling will come not from more impoundments but from confronting demand and allocation, the harder and unavoidable adaptation.

5. Data and code availability

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.XXXXXXX). The pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (standardized drought indices), MODIS MOD16/MOD13/MOD17 (evapotranspiration and vegetation), MapBiomas Chile (land cover), the CR2 network (streamflow), SRTM (elevation), and the DGA reservoir and watershed registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

References

Alvarez-Garreton, C., Boisier, J.P., Garreaud, R., Seibert, J., Vis, M., 2021. Progressive water deficits during multiyear droughts in basins with long hydrological memory in Chile. Hydrology and Earth System Sciences 25, 429–446. URL: https://hess.copernicus.org/articles/25/429/2021/, doi:10.5194/hess-25-429-2021.

Barría, P., Sandoval, I.B., Guzman, C., Chadwick, C., Alvarez-Garreton, C., Díaz-Vasconcellos, R., Ocampo-Melgar, A., Fuster, R., 2021. Water allocation under climate change. Elementa: Science of the Anthropocene 9, 00131. URL: https://online.ucpress.edu/elementa/article/9/1/00131/117183/Water-allocation-under-climate-changeA-diagnosis, doi:10.1525/elementa.2020.00131.

Condon, L.E., Atchley, A.L., Maxwell, R.M., 2020. Evapotranspiration depletes groundwater under warming over the contiguous United States. Nature Communications 11, 873. URL: https://www.nature.com/articles/s41467-020-14688-0, doi:10.1038/s41467-020-14688-0.

Di Baldassarre, G., Wanders, N., AghaKouchak, A., Kuil, L., Rangecroft, S., Veldkamp, T.I.E., Garcia, M., Van Oel, P.R., Breinl, K., Van Loon, A.F., 2018. Water shortages worsened by reservoir effects. Nature Sustainability 1, 617–622. URL: https://www.nature.com/articles/s41893-018-0159-0, doi:10.1038/s41893-018-0159-0.

Garreaud, R., Boisier, J.P., Álvarez Garreton, C., Christie, D., Carrasco-Escaff, T., Vergara, I., Chávez, R.O., Aldunce, P., Camus, P., Suazo-Álvarez, M., Masiokas, M., Castro, G., Muñoz, A., Zambrano-Bigiarini, M., Fuster, R., Godoy, L., 2025. Hyperdroughts in central Chile: Drivers, Impacts and Projections. URL: https://egusphere.copernicus.org/preprints/2025/egusphere-2025-517/, doi:10.5194/egusphere-2025-517.

Garreaud, R.D., Alvarez-Garreton, C., Barichivich, J., Boisier, J.P., Christie, D., Galleguillos, M., LeQuesne, C., McPhee, J., Zambrano-Bigiarini, M., 2017. The 2010–2015 megadrought in central Chile: impacts on regional hydroclimate and vegetation. Hydrology and Earth System Sciences 21, 6307–6327. URL: https://hess.copernicus.org/articles/21/6307/2017/, doi:10.5194/hess-21-6307-2017.

Glade, F.E., Miranda, M.D., Meza, F.J., van Leeuwen, W.J.D., 2016. Productivity and phenological responses of natural vegetation to present and future inter-annual climate variability across semi-arid river basins in Chile. Environmental Monitoring and Assessment 188, 676. URL: http://dx.doi.org/10.1007/s10661-016-5675-7, doi:10.1007/s10661-016-5675-7.

Grafton, R.Q., Williams, J., Perry, C.J., Molle, F., Ringler, C., Steduto, P., Udall, B., Wheeler, S.A., Wang, Y., Garrick, D., Allen, R.G., 2018. The paradox of irrigation efficiency. Science 361, 748–750. doi:10.1126/science.aat9314. publisher: American Association for the Advancement of Science.

Kellner, E., 2021. The controversial debate on the role of water reservoirs in reducing water scarcity. WIREs Water 8, e1514. URL: https://wires.onlinelibrary.wiley.com/doi/10.1002/wat2.1514, doi:10.1002/wat2.1514.

Lee, C., Kang, S., Kim, S., 2026. Drought resistance of dams based on propagation analysis: A case study of various multipurpose dams in South Korea. Journal of Hydrology: Regional Studies 63, 103106. URL: https://linkinghub.elsevier.com/retrieve/pii/S2214581826000042, doi:10.1016/j.ejrh.2026.103106.

Lema, F., Mendoza, P.A., Vásquez, N.A., Mizukami, N., Zambrano-Bigiarini, M., Vargas, X., 2025. Technical note: What does the Standardized Streamflow Index actually reflect? Insights and implications for hydrological drought analysis. Hydrology and Earth System Sciences 29, 1981–2002. URL: https://hess.copernicus.org/articles/29/1981/2025/, doi:10.5194/hess-29-1981-2025.

Li, Y., Zhao, G., Allen, G.H., Gao, H., 2023. Diminishing storage returns of reservoir construction. Nature Communications 14, 3203. URL: https://www.nature.com/articles/s41467-023-38843-5, doi:10.1038/s41467-023-38843-5.

López-Moreno, J.I., Vicente-Serrano, S.M., Beguería, S., García-Ruiz, J.M., Portela, M.M., Almeida, A.B., 2009. Dam effects on droughts magnitude and duration in a transboundary basin: The Lower River Tagus, Spain and Portugal. Water Resources Research 45, 2008WR007198. URL: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2008WR007198, doi:10.1029/2008WR007198.

Macpherson, E., Clark, C., Salazar, P.W., Baird, N., Akhtar-Khavari, A., Challies, E., 2023. Evolving rights to (and of) water in Chile: a case for relationship-based water law and governance. The International Journal of Human Rights , 1–26URL: https://www.tandfonline.com/doi/full/10.1080/13642987.2023.2266719, doi:10.1080/13642987.2023.2266719.

McKee, T.B., Doesken, N.J., Kleist, J., 1993. The relationship of drought frequency and duration to time scales. Proceedings of the Ninth Conference on Applied Climatology, American Meteorological Society 17, 179–184.

Meza, F.J., 2013. Recent trends and {ENSO} influence on droughts in {N}orthern {C}hile: {A}n application of the {S}tandarized {P}recipitation {E}vapotranspiration {I}ndex. Weather and Climate Extremes 1, 51–58.

Oertel, M., Meza, F.J., Gironás, J., 2020. Observed trends and relationships between ENSO and standardized hydrometeorological drought indices in central Chile. Hydrological Processes 34, 159–174. URL: https://onlinelibrary.wiley.com/doi/10.1002/hyp.13596, doi:10.1002/hyp.13596.

Reade Malagueño, B., D'Odorico, P., 2024. *¿Libre de la Maleza Estatista?* Assessing Neoliberal Promises and Water Markets in Chile. Earth's Future 12, e2024EF004963. URL: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024EF004963, doi:10.1029/2024EF004963.

Rivera-Vidal, R., Arumí, J.L., Melo, O., Delgado, V., Parra, V., Stehr, A., Daniele, L., 2025. Managed aquifer recharge implementation challenges: Lessons from Chile's water-scarce regions. Groundwater for Sustainable Development 31, 101502. URL: https://linkinghub.elsevier.com/retrieve/pii/S2352801X25000992, doi:10.1016/j.gsd.2025.101502.

Scanlon, B.R., Fakhreddine, S., Rateb, A., De Graaf, I., Famiglietti, J., Gleeson, T., Grafton, R.Q., Jobbagy, E., Kebede, S., Kolusu, S.R., Konikow, L.F., Long, D., Mekonnen, M., Schmied, H.M., Mukherjee, A., MacDonald, A., Reedy, R.C., Shamsudduha, M., Simmons, C.T., Sun, A., Taylor, R.G., Villholth, K.G., Vörösmarty, C.J., Zheng, C., 2023. Global water resources and the role of groundwater in a resilient water future. Nature Reviews Earth & Environment 4, 87–101. URL: https://www.nature.com/articles/s43017-022-00378-6, doi:10.1038/s43017-022-00378-6.

Schilstra, M., Wang, W., Van Oel, P.R., Wang, J., Cheng, H., 2024. The effects of reservoir storage and water use on the upstream–downstream drought propagation. Journal of Hydrology 631, 130668. URL: https://linkinghub.elsevier.com/retrieve/pii/S0022169424000623, doi:10.1016/j.jhydrol.2024.130668.

Venegas-González, A., Muñoz, A.A., Carpintero-Gibson, S., González-Reyes, A., Schneider, I., Gipolou-Zuñiga, T., Aguilera-Betti, I., Roig, F.A., 2023. Sclerophyllous Forest Tree Growth Under the Influence of a Historic Megadrought in the Mediterranean Ecoregion of Chile. Ecosystems 26, 344–361. URL: https://link.springer.com/10.1007/s10021-022-00760-x, doi:10.1007/s10021-022-00760-x.

Vicente-Serrano, S.M., Beguería, S., López-Moreno, J.I., 2010. A multiscalar drought index sensitive to global warming: The standardized precipitation evapotranspiration index. Journal of Climate 23, 1696–1718. URL: http://dx.doi.org/10.1175/2009JCLI2909.1, doi:10.1175/2009JCLI2909.1.

Vicente-Serrano, S.M., López-Moreno, J.I., Beguería, S., Lorenzo-Lacruz, J., Azorin-Molina, C., Morán-Tejeda, E., 2012. Accurate computation of a streamflow drought index. J. Hydrol. Eng. 17(2), 318–332. URL: http://hdl.handle.net/10261/49224.

Vicente-Serrano, S.M., Peña-Angulo, D., Beguería, S., Domínguez-Castro, F., Tomás-Burguera, M., Noguera, I., Gimeno-Sotelo, L., El Kenawy, A., 2022. Global drought trends and future projections. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences 380, 20210285. URL: https://royalsocietypublishing.org/doi/10.1098/rsta.2021.0285, doi:10.1098/rsta.2021.0285.

Wang, Z., Jiang, L., Nielsen, K., Wang, L., 2024. Reservoir Filling Up Problems in a Changing Climate: Insights From CryoSat-2 Altimetry. Geophysical Research Letters 51, e2024GL108934. URL: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL108934, doi:10.1029/2024GL108934.

Wu, J., Chen, X., Yao, H., Gao, L., Chen, Y., Liu, M., 2017. Non-linear relationship of hydrological drought responding to meteorological drought and impact of a large reservoir. Journal of Hydrology 551, 495–507. URL: https://linkinghub.elsevier.com/retrieve/pii/S002216941730433X, doi:10.1016/j.jhydrol.2017.06.029.

Wu, J., Liu, Z., Yao, H., Chen, X., Chen, X., Zheng, Y., He, Y., 2018. Impacts of reservoir operations on multi-scale correlations between hydrological drought and meteorological drought. Journal of Hydrology 563, 726–736. URL: https://linkinghub.elsevier.com/retrieve/pii/S0022169418304761, doi:10.1016/j.jhydrol.2018.06.053.

Zambrano, F., Vrieling, A., Meza, F., Duran-Llacer, I., Fernández, F., Venegas-González, A., Raab, N., Craven, D., 2025. From Drought to Aridification: Land-Cover Fingerprints of a Drying Chile. Earth's Future 13, e2025EF006744. URL: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025EF006744, doi:10.1029/2025EF006744.

Zambrano, F., Vrieling, A., Nelson, A., Meroni, M., Tadesse, T., 2018. Prediction of drought-induced reduction of agricultural productivity in Chile from MODIS, rainfall estimates, and climate oscillation indices. Remote Sensing of Environment 219, 15–30. URL: https://www.sciencedirect.com/science/article/pii/S0034425718304541, doi:10.1016/j.rse.2018.10.006. publisher: Elsevier.

Zambrano, F., Wardlow, B., Tadesse, T., Lillo-Saavedra, M., Lagos, O., 2017. Evaluating satellite-derived long-term historical precipitation datasets for drought monitoring in Chile. Atmospheric Research 186, 26–42. URL: https://www.sciencedirect.com/science/article/pii/S0169809516305865, doi:10.1016/j.atmosres.2016.11.006. publisher: Elsevier.